COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S1
DATE PERFORMED:	07/09/2026	**DATE SUBMITTED:**	07/14/26
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ACTIVITY NO. 1 **[ACTIVITY NAME] Hands-on Activity 1.1 Basic C++ Programming**

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

A function prototype in c++ is a declaration of the name, parameters and return type of the function before its definition. Write a C++ code the satisfies the following:

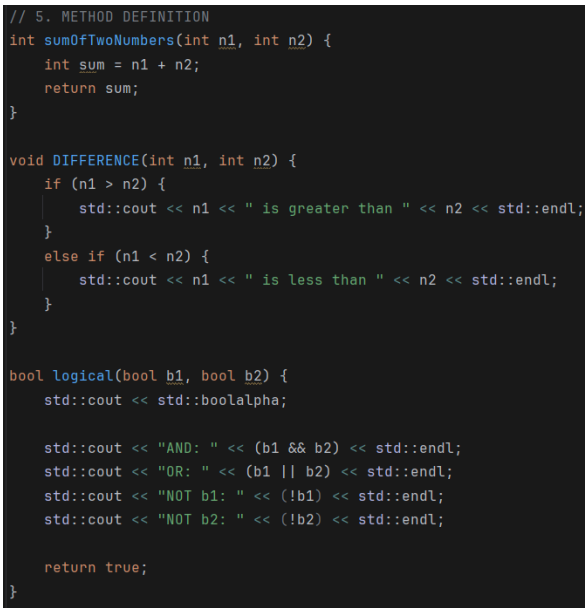
- Create a function that will take two numbers and display the sum.
- Create a function that will return whether variable A is greater than variable B.
- Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

Header File Declaration Section	`// 1. HEADER FILE DECLARATION SECTION` `#include <iostream>`

Global Declaration Section	`// 2. GLOBAL DECLARATION SECTION`
	`int sumOfTwoNumbers(int n1, int n2);`
	`void DIFFERENCE(int n1, int n2);`
	`bool logical(bool b1, bool b2);`

Class Declaration and Method Definition Section	`// 3. CLASS DECLARATION AND METHOD DEFINITION SECTION` `// (Left empty since this specific program is procedural and doesn't utilize OOP classes)`

Main Function	`// 4. MAIN FUNCTION`
	`int main() {`
	`  std::cout << sumOfTwoNumbers(3, 4) << std::endl;`
	`  DIFFERENCE(3, 4);`
	`  logical(true, false);`
	`  return 0;`
	`}`

Method Definition	

```cpp
// 5. METHOD DEFINITION
int sumOfTwoNumbers(int n1, int n2) {
    int sum = n1 + n2;
    return sum;
}

void DIFFERENCE(int n1, int n2) {
    if (n1 > n2) {
        std::cout << n1 << " is greater than " << n2 << std::endl;
    }
    else if (n1 < n2) {
        std::cout << n1 << " is less than " << n2 << std::endl;
    }
}

bool logical(bool b1, bool b2) {
    std::cout << std::boolalpha;

    std::cout << "AND: " << (b1 && b2) << std::endl;
    std::cout << "OR: " << (b1 || b2) << std::endl;
    std::cout << "NOT b1: " << (!b1) << std::endl;
    std::cout << "NOT b2: " << (!b2) << std::endl;

    return true;
}
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements



```
"C:\Users\Bienvenido\CLionProjects\C++\PRELIM (2)\ILO_B.exe"
The shape is a valid Triangle


Process finished with exit code 0
```

The program only validates angle sums, deliberately skipping area, perimeter, and user input. Its core flaw is using total angle of less than equal to 180 instead of equal to 180, which lets impossible triangles pass as valid.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

1. Create a C++ program to swap the two numbers in different variables.

```cpp
#include<iostream>

int main() {
    int a = 6, b = 7;

    std::cout << a << b << std::endl;
    std:: swap( [&] a, [&] b);
    std:: cout << a << b << std::endl;
}
```

```
main
Run  PRELIM__3_
"C:\Users\Bienvenido\CLionProjects\C++\PR
67
76

Process finished with exit code 0
```

2. Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

```cpp
#include<iostream>

double kelvinToFahrenheit(double kelvin) {
    double celsius = kelvin - 273.15;
    double fahrenheit = celsius * 9.0 / 5.0 + 32.0;
    return fahrenheit;
}

int main() {
    double kelvin;

    std::cout << "Enter kelvin: ";
    std::cin >> kelvin;

    if (kelvin < 0) {
        std::cout << "Invalid input: temperature cannot be below absolute ze
        return 1;
    }

    double fahrenheit = kelvinToFahrenheit(kelvin);
```

```
ain
PRELIM_3_2_
"C:\Users\Bienvenido\CLionProjects\C++\PRELIM (3.2)\cmake-build-debug\PRELIM_
Enter kelvin:290

290 K: 62.33 F
```

3. Create a C++ program that has a function that will calculate the distance between two points.

```cpp
#include<iostream>

double kelvinToFahrenheit(double kelvin) {
    double celsius = kelvin - 273.15;
    double fahrenheit = celsius * 9.0 / 5.0 + 32.0;
    return fahrenheit;
}

int main() {
    double kelvin;

    std::cout << "Enter kelvin: ";
    std::cin >> kelvin;

    if (kelvin < 0) {
        std::cout << "Invalid input: cannot be zero (0 K).\n";
        return 1;
    }

    double fahrenheit = kelvinToFahrenheit(kelvin);
```

```
ain
PRELIM__3_2_
"C:\Users\Bienvenido\CLionProjects\C++\PRELIM (3.2)\cmake-build-
Enter kelvin:290

290 K: 62.33 F

Process finished with exit code 0
```

4. Modify the code given in ILO B and add the following functions:
a. A function to compute for the area of a triangle
b. A function to compute for the perimeter of a triangle
c. A function that determines whether the triangle is acute-angled, obtuse-angled or 'others.'

```cpp
#include <iostream>
#include <cmath>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;
    double sideA, sideB, sideC;
public:
    Triangle(double A, double B, double C);
    void getAngles(double A, double B, double C);
    void getSides(double a, double b, double c);
    const bool validateTriangle();
    double computeArea();
    double computePerimeter();
    std::string classifyTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
    sideA = sideB = sideC = 0;
}

void Triangle::getAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::getSides(double a, double b, double c) {
    sideA = a;
    sideB = b;
    sideC = c;
}

const bool Triangle::validateTriangle() {
    return (std::abs(totalAngle - 180) < 1e-6);
}

double Triangle::computeArea() {
    double s = (sideA + sideB + sideC) / 2.0;
    return std::sqrt(s * (s - sideA) * (s - sideB) * (s - sideC));
}

double Triangle::computePerimeter() {
    return sideA + sideB + sideC;
}

std::string Triangle::classifyTriangle() {
    // find the largest angle without std::max
    double maxAngle = angleA;
    if (angleB > maxAngle) maxAngle = angleB;
    if (angleC > maxAngle) maxAngle = angleC;

    if (std::abs(maxAngle - 90) < 1e-6) return "Others (right-angled)";
    else if (maxAngle > 90) return "Obtuse-angled";
    else return "Acute-angled";
}
```

```
    else return "Acute-angled";
}

int main() {
    double a, b, c, s1, s2, s3;

    std::cout << "Enter angle A, B, C: ";
    std::cin >> a >> b >> c;

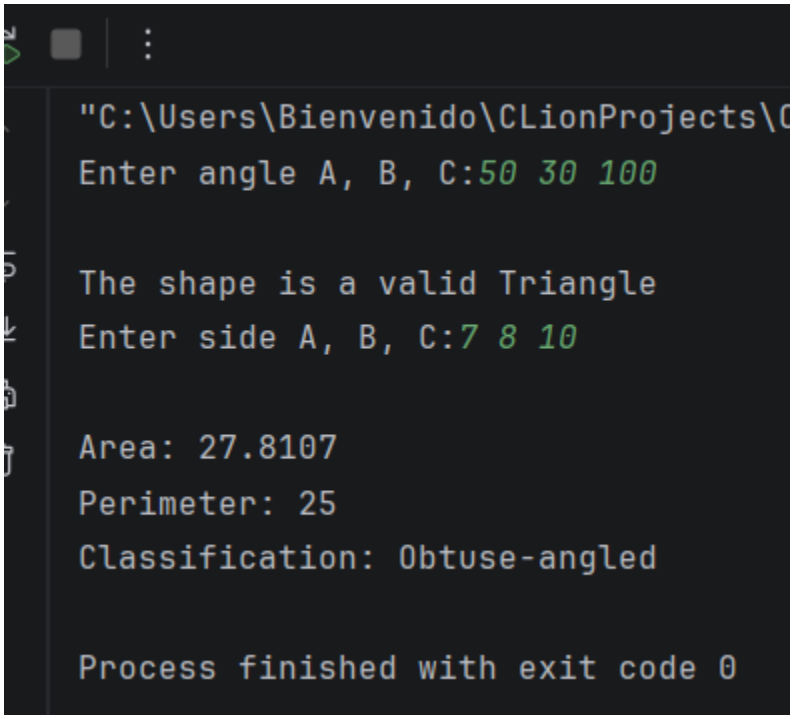
    Triangle set1(a, b, c);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid Triangle" << std::endl;

        std::cout << "Enter side A, B, C: ";
        std::cin >> s1 >> s2 >> s3;
        set1.getSides( a: s1,  b: s2,  c: s3);

        std::cout << "Area: " << set1.computeArea() << std::endl;
        std::cout << "Perimeter: " << set1.computePerimeter() << std::endl;
        std::cout << "Classification: " << set1.classifyTriangle() << std::en
    }

    return 0;
}
```



Conclusion
Provide the following:

Summary of lessons learned

- In this activity, I learned the importance of proper program structure in C++ from header declarations down to class methods.
-

Analysis of the procedure

- I learned how to properly organize a C++ program by separating function prototypes from their actual definitions to improve code readability.

Analysis of the supplementary activity

- Through the supplementary activities, I successfully wrote c++ functions to swap variables and convert temperatures. I also managed to update the triangle class to calculate its area, perimeter, and specific type.
-

Concluding statement / Feedback: How well did you think you did in this activity? What are your areas for improvement?

- Although I managed to finish all the tasks, I am still having a hard time because I am quite a little forgetful (makalimutin) when it comes to remembering all the strict rules of c++. Since I easily forget the exact syntax, like typing semicolons, declaring strict data types, and setting up function prototypes, I still need more adjustments transitioning from python's simpler style.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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